

Liquids and Solids

Chapter Outline

[10.1 Intermolecular Forces](#)

[10.2 Properties of Liquids](#)

[10.3 Phase Transitions](#)

[10.4 Phase Diagrams](#)

[10.5 The Solid State of Matter](#)

[10.6 Lattice Structures in Crystalline Solids](#)

CH 10.2 I made a concept video, but it's optional
(you will not be tested on this content)

Sections 10.5/10.6 Will not be covered in CHEM 120.



Marie-Claude Paquette throwing the boiling water into the -30 centigrade air creating beautiful designs (St. Agathe, QC)

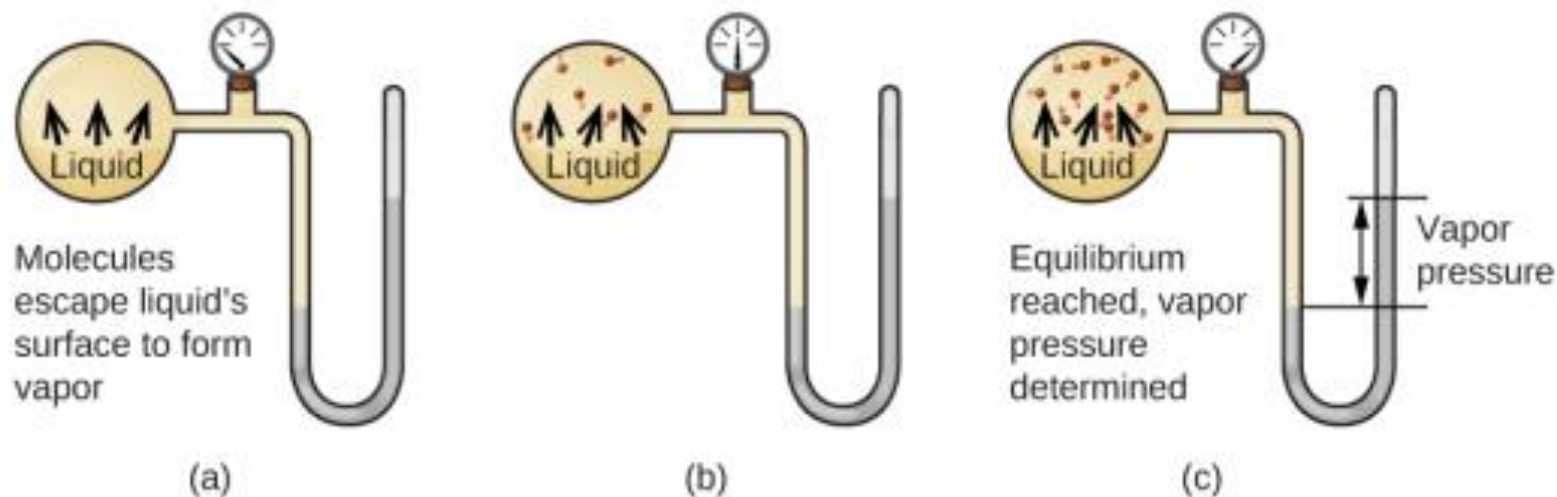


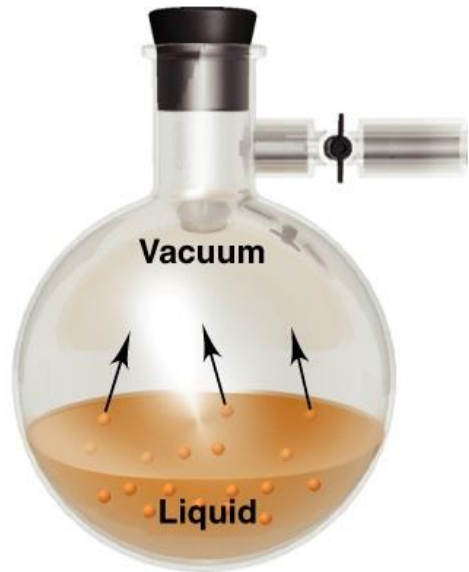
Figure 10.22 In a closed container, dynamic equilibrium is reached when (a) the rate of molecules escaping from the liquid to become the gas (b) increases and eventually (c) equals the rate of gas molecules entering the liquid. When this equilibrium is reached, the vapor pressure of the gas is constant, although the vaporization and condensation processes continue.

The pressure exerted by the vapor in equilibrium with a liquid in a closed container at a given temperature is called the liquid's **vapor pressure** (or equilibrium vapor pressure).

The chemical identities of the molecules in a liquid determine the types (and strengths) of intermolecular attractions possible; consequently, different substances will exhibit different equilibrium vapor pressures at the same temperature.

Vapor Pressure via Kinetic Molecular theory

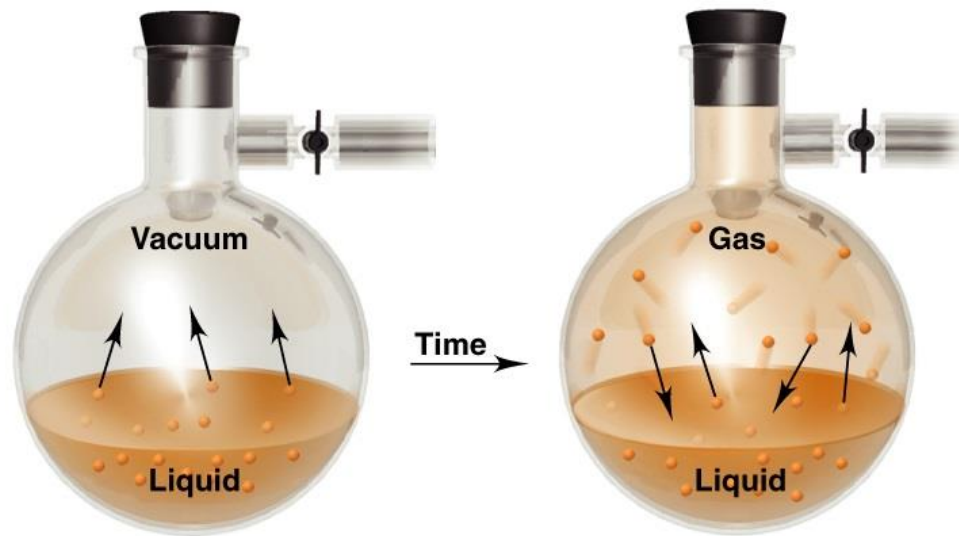
- Suppose we place some liquid in a sealed chamber.
- Molecules at the surface of the liquid will undergo evaporation.



A Molecules in the liquid vaporize.

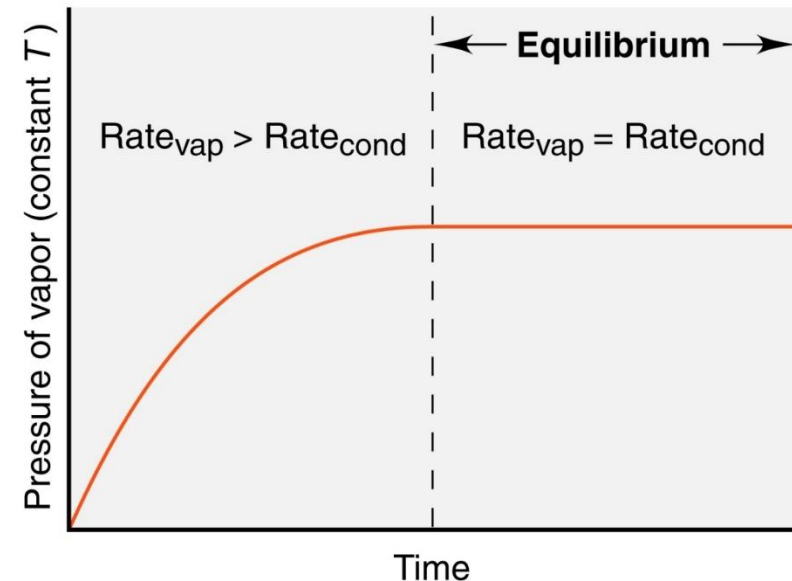
Vapor Pressure via Kinetic Molecular theory

- Liquid phase molecules evaporate.
- Vapor phase molecules condense.
- Processes are simultaneous.
 - Dynamic equilibrium



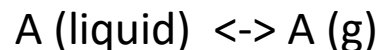
A Molecules in the liquid vaporize.

B Molecules vaporize and condense at the same rate.

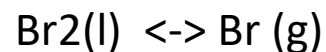


C Plot of pressure vs. time.

Whenever liquid is placed in a closed container and allowed to equilibrate, as long as some liquid remains then the vapor above the liquid will be at the **vapor pressure**.



$$K = p_A = \text{vapor pressure}$$



$$K = 0.3 \text{ bar (at 298 K)}$$

It is common to discuss vapor pressure rather than the equilibrium constants for phase changes, So to get this info you would look up the vapor pressure of Bromine in tables, it would not be listed as a table of K values.

This is because the **vapor pressure is simply the equilibrium constant for the vaporization process. However, it is commonly reported in units other than bar, such as mmHg or torr.**

Volatility



- Volatility is a property/description based on the vapor pressure.
- High Vapor Pressure – Volatile
 - evaporate easily
 - acetone, ether, gasoline, ...

Volatility



- Low Vapor Pressure – Nonvolatile
 - does not evaporate easily
 - oil, pitch, ...

Vapour Pressure & Temperature

This we can also understand by appealing to kinetic molecular theory, or thermodynamics.
We'll start with kinetic molecular theory....

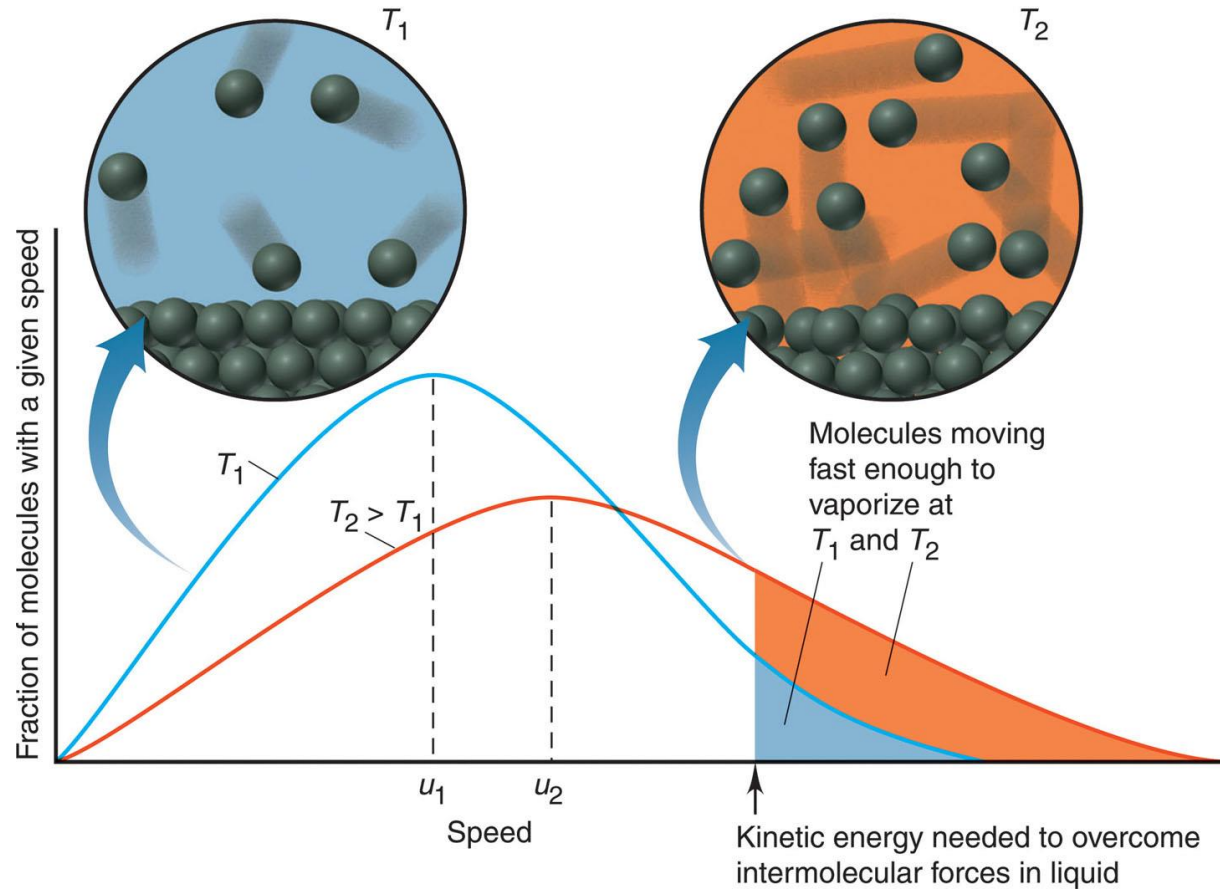
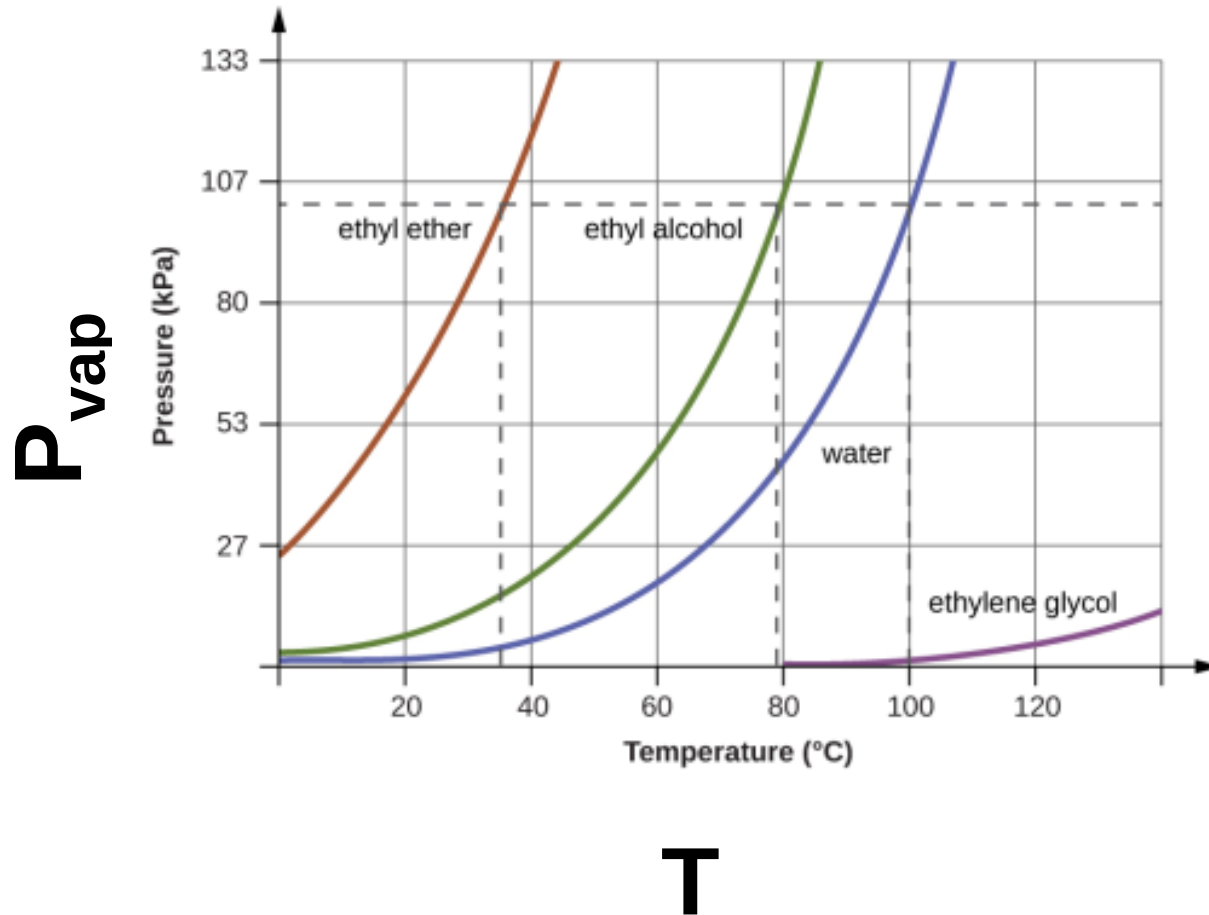


Figure 11.6

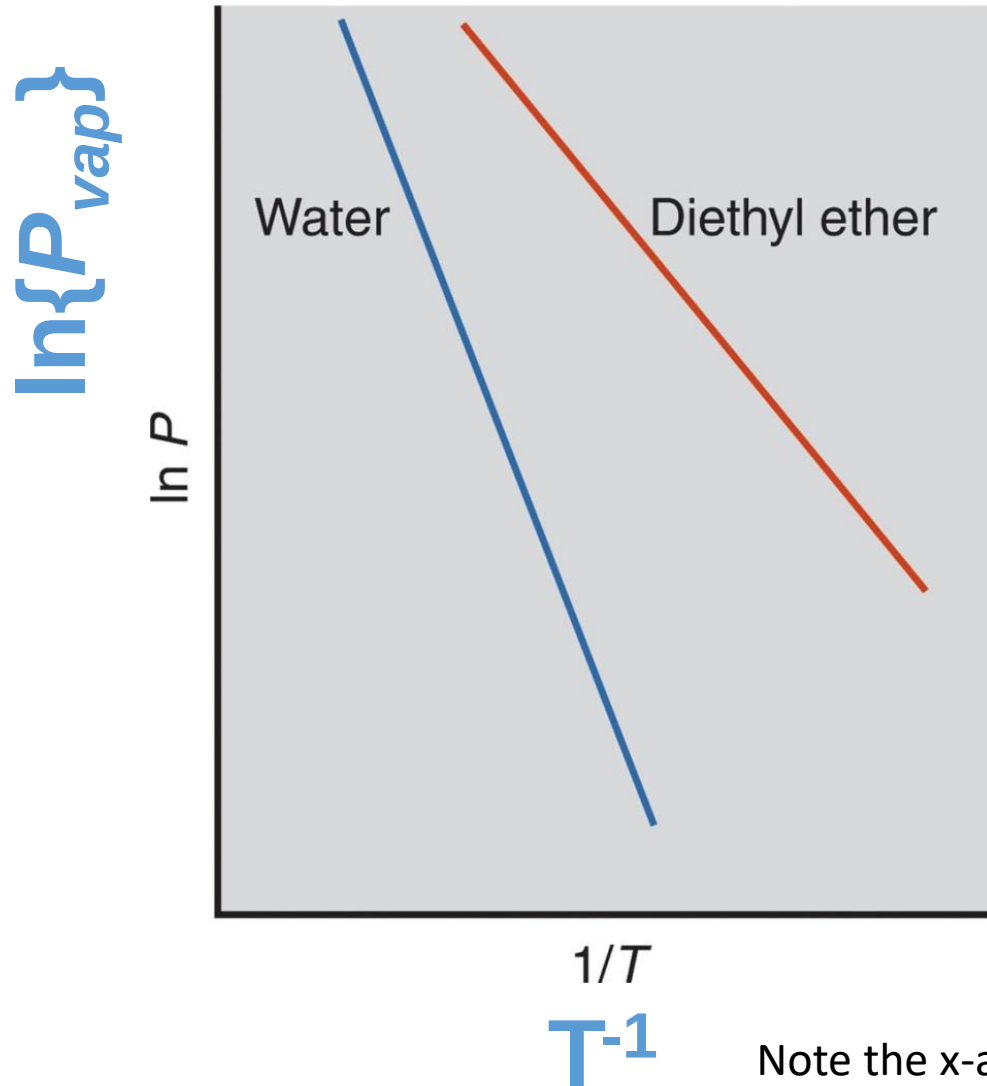
Vapor Pressure



- We know the process is endothermic (breaking intermolecular forces), so K will increase with Temp.
- Indeed Vapor pressure increases with increasing temperature.
- Vapor Pressure decreases as strength of intermolecular forces increases
- Relationship between P_{vap} and T is non-linear.

$$P = Ae^{-\Delta H_{vap}/RT}$$

Vapor Pressure

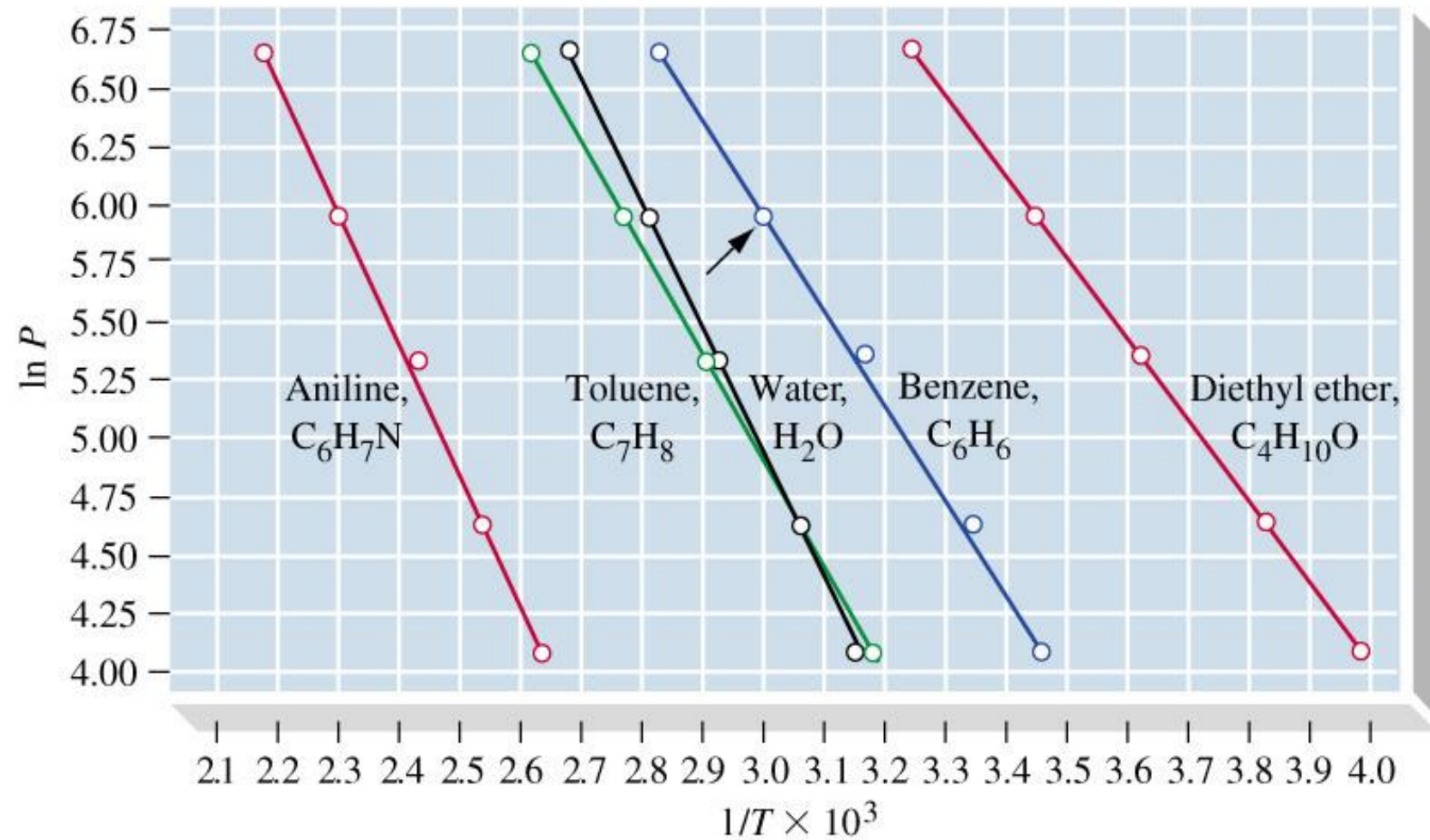


- However, as we saw with van't Hoff plots, we can fix this.
- Plot of $\ln\{P_{vap}\}$ vs T^{-1} is linear.

$$\ln P = -\frac{\Delta H_{vap}}{RT} + \ln A$$

Note the x-axis is inverse Temperature
($1/T$) = T^{-1}

Clausius-Clapeyron Equation



Clausius-Clapeyron Equation

$$\ln \left\{ \frac{P_2}{P_1} \right\} = - \frac{\Delta H_{vap}}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right) \quad \text{or} \quad \ln \left(\frac{P_2}{P_1} \right) = \frac{\Delta H_{vap}}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

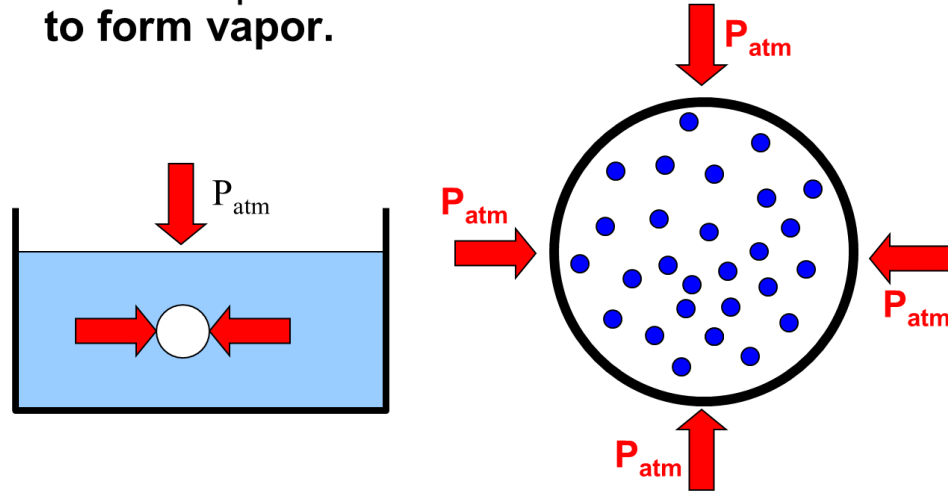
Dr. Sewall's Preferred Version

- Relates enthalpy of vaporization with change in temperature and change in vapor pressure.
- Just a specialized version of the van't Hoff equation!

Boiling Point

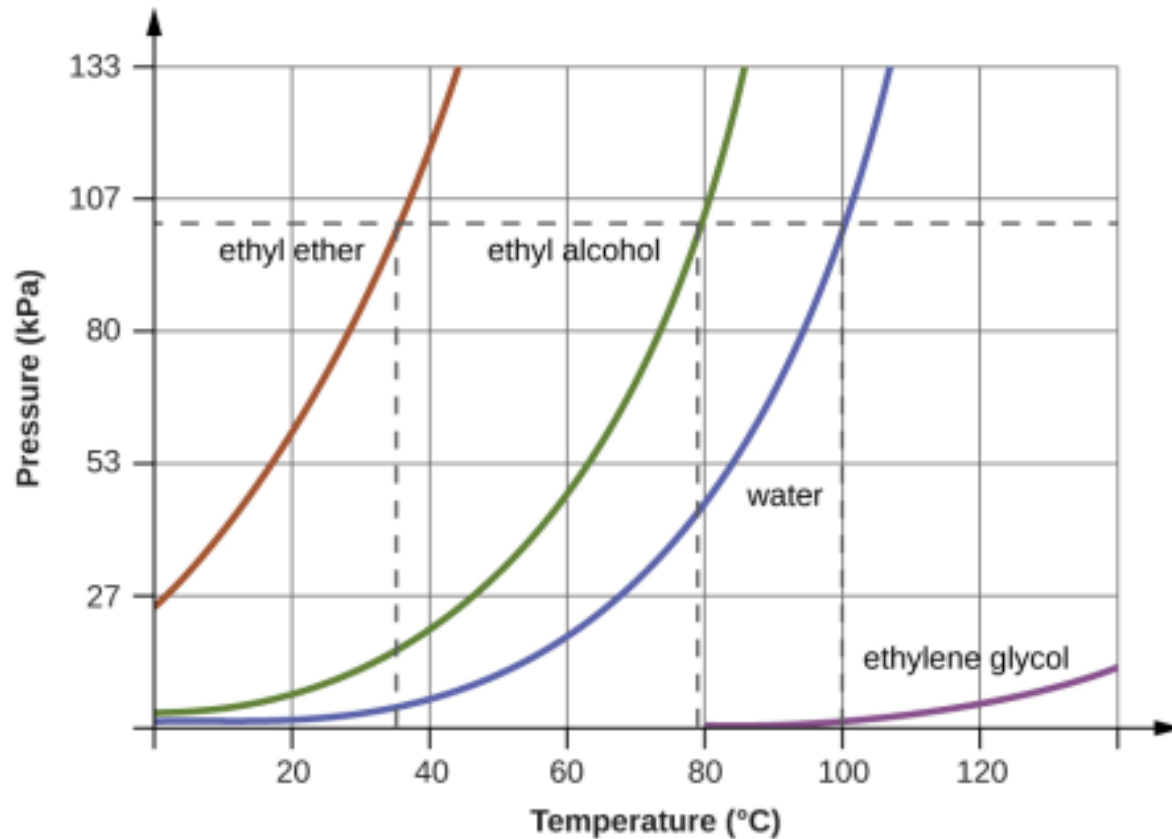


- At about 100°C, water bubbles vigorously and is rapidly converted to water vapor.
 - When $P_{\text{vap}} \geq P_{\text{atm}}$ bulk liquid can expand to form vapor.



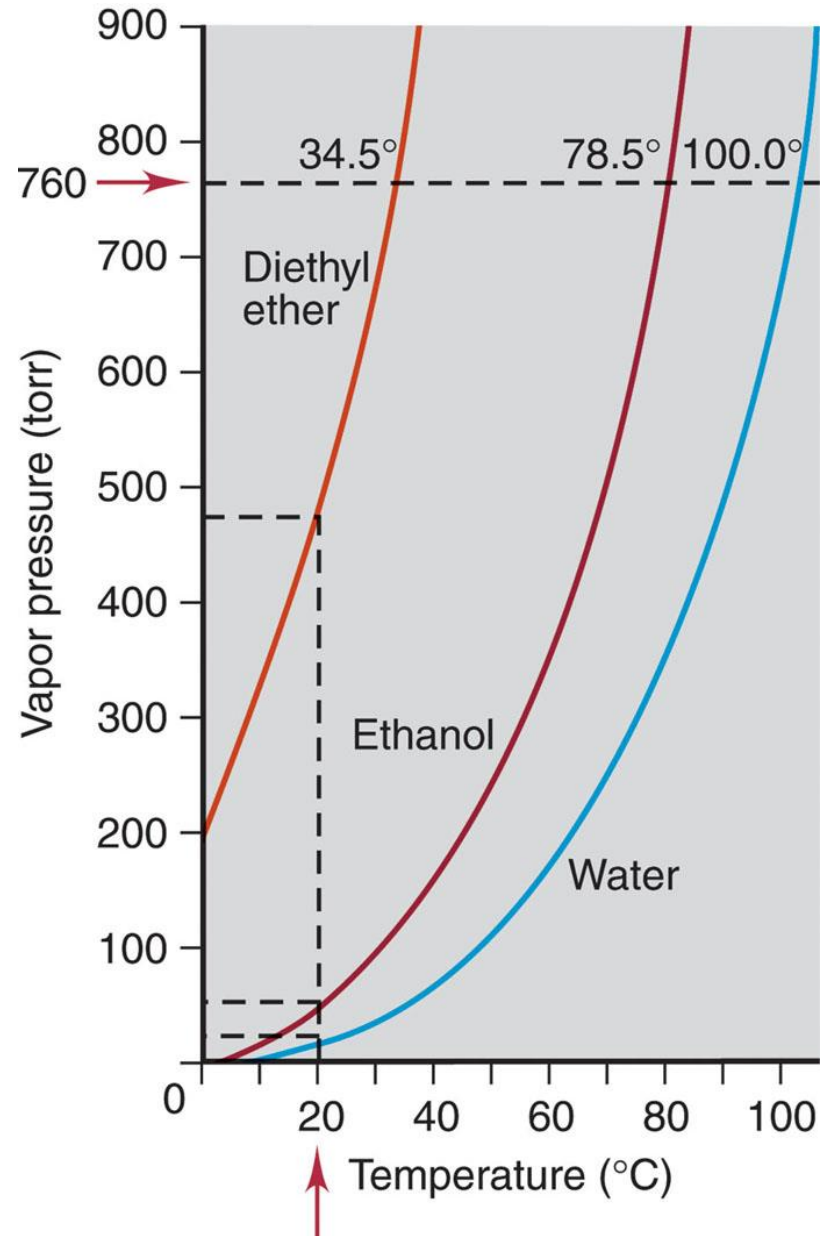
- The **boiling point** of a liquid is the temperature at which the vapor pressure equals the external pressure. The liquid phase is not stable (observable at equilibrium) above the boiling point.
- The **normal boiling point** of a substance is observed at standard atmospheric pressure or 760 torr (1 bar).
- As the external pressure on a liquid **increases**, the boiling point **increases**.

Boiling Point

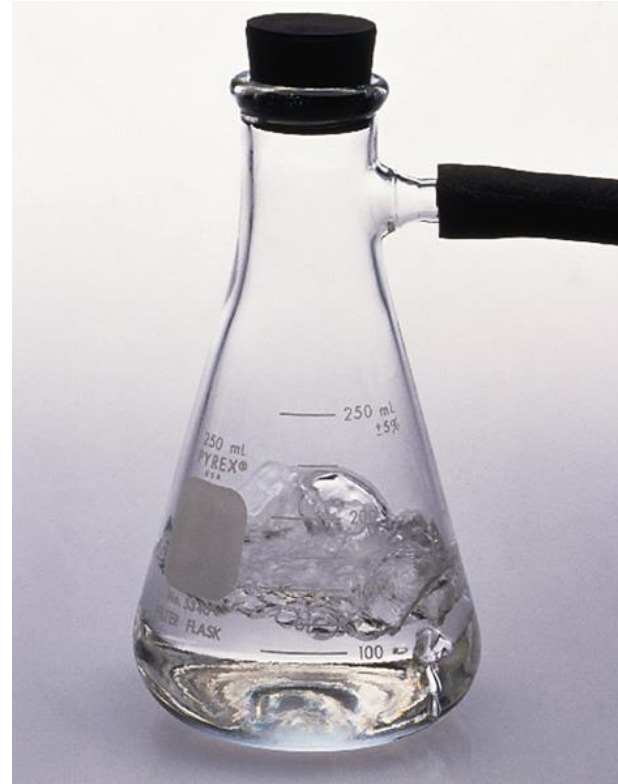


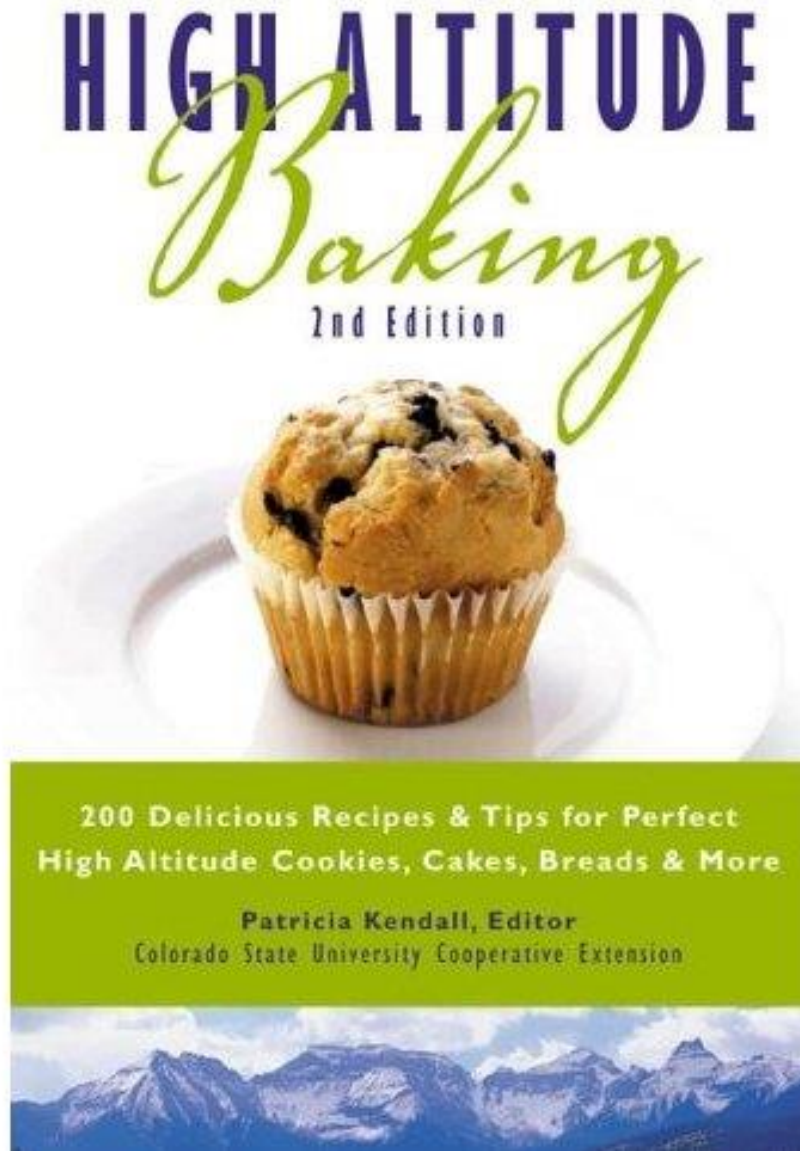
- H₂O vapor pressure curve intersects atmospheric pressure at 100°C
- This is the *normal* boiling point of water. $P_{\text{vap}} = P_{\text{atm}}$

Boiling Point



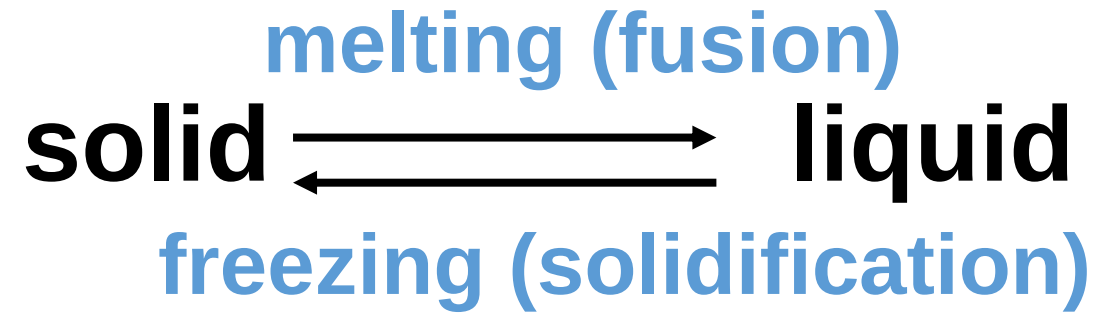
- If you could reduce pressure to less than 24 torr (.03 atm), water would boil at room temperature.





- At high altitudes, atmospheric pressure is reduced.
- Water boils at a slightly lower temperature (approx. 95°C in Denver, Colorado)

10.3 Phase Transitions



During melting (fusion), a molecule goes from an environment with strong, specific interactions (solid), to one with weakened, more dynamic interactions (liquid).



Enthalpy of Fusion

$$\Delta H_{\text{fusion}} = H_{\text{liquid}} - H_{\text{solid}}$$

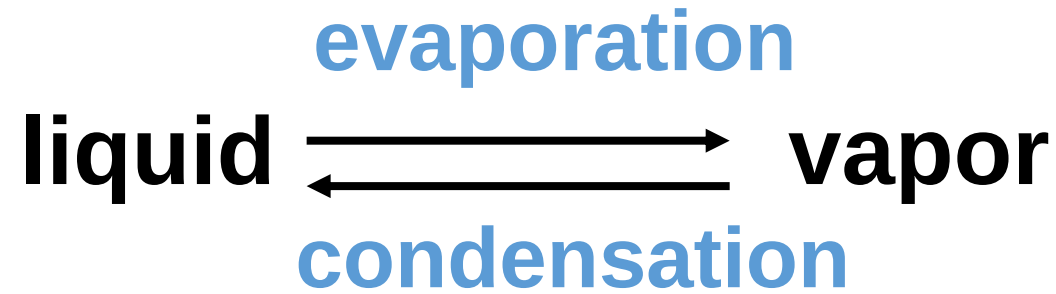
- $H_{\text{liquid}} > H_{\text{solid}}$
- Enthalpy of fusion is +ve.
 - melting is endothermic.
 - freezing is exothermic.
- It takes energy to break the strong interactions that hold together a solid.

Enthalpy of Fusion



- Not just because ice cubes are cold.
- Melting is endothermic
 - energy goes into breaking hydrogen bonds in ice.

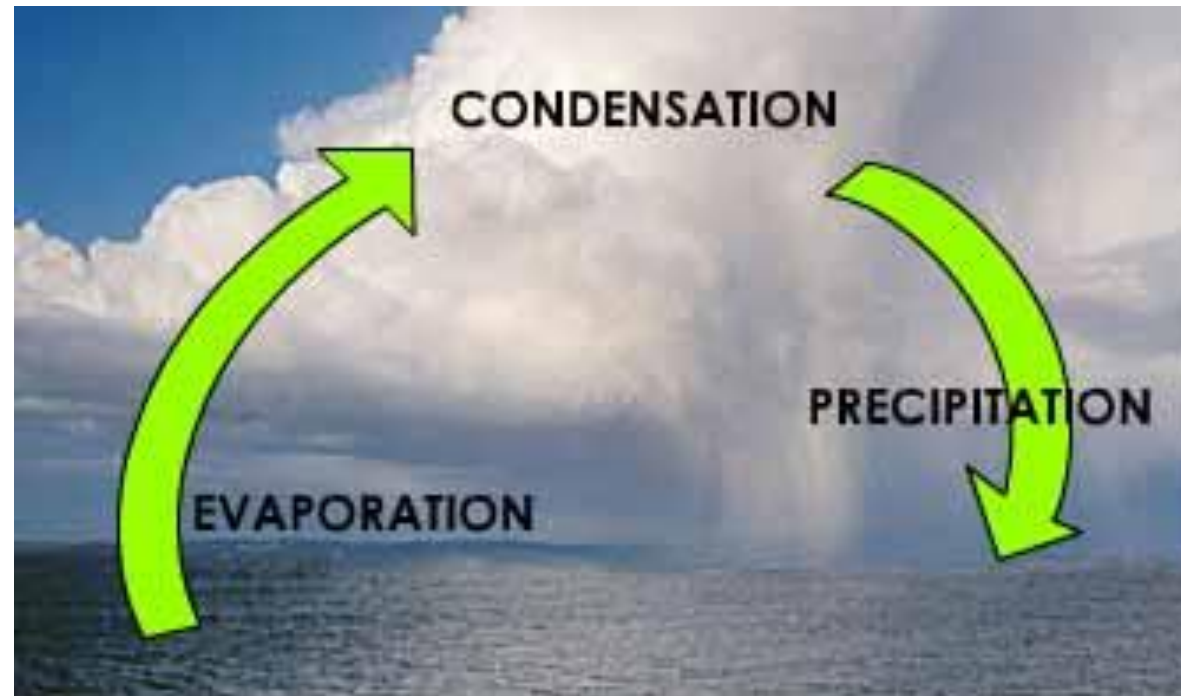
Phase Changes: Liquid to Gas



During evaporation, a molecule goes from an environment with many attractive interactions (liquid), to one with few interactions (gas).

The heat exchanged during these processes can be measured, known as the enthalpy of vaporization (ΔH_{vap}) or condensation (ΔH_{cond}).

It is easily predictable which process is exothermic or endothermic.



Enthalpy of Vaporization



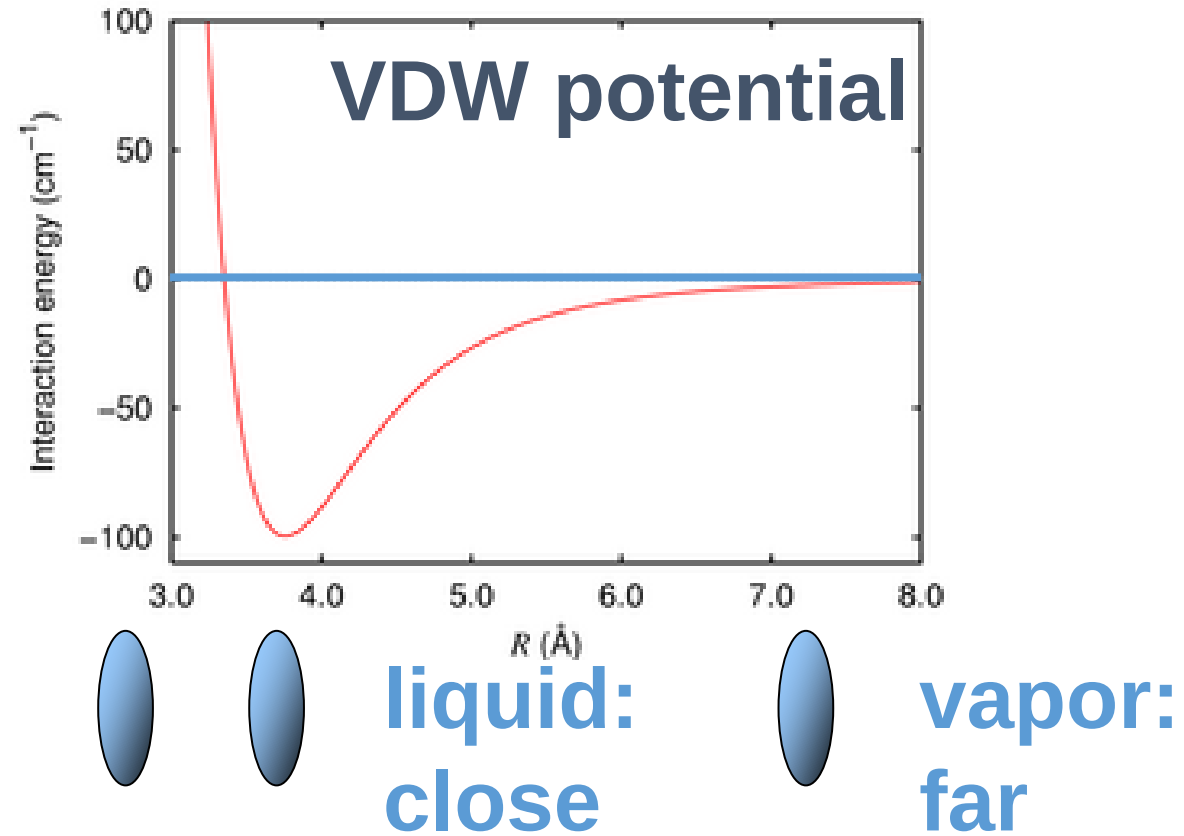
- It takes energy to break attractive interactions between molecules.
- Heat is absorbed from the surroundings when liquids evaporate.
- Vaporization is endothermic.

Enthalpy of Condensation



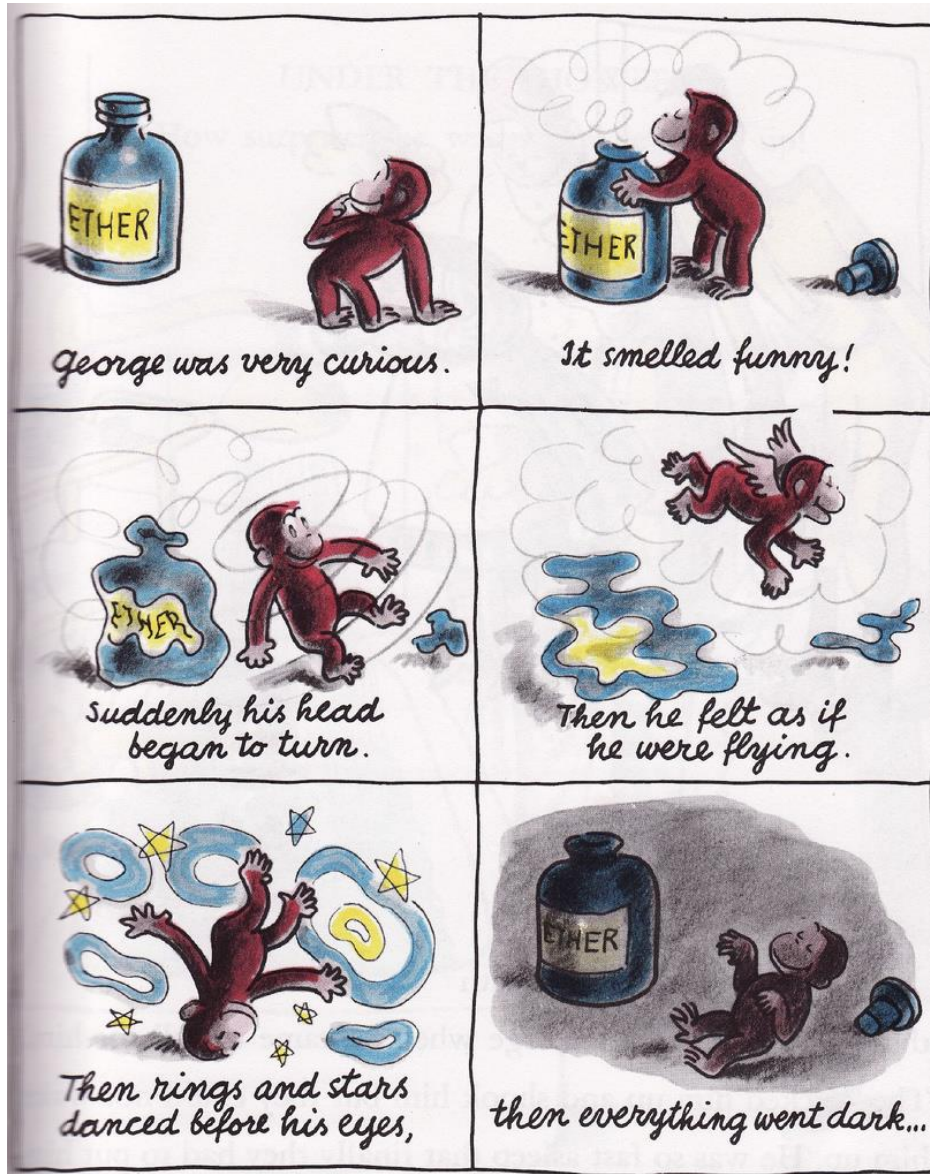
- Attractive interactions are formed as a molecule enters the liquid state.
- Heat is released when gases condense to form liquids.
- Condensation is exothermic.

Enthalpy of Vaporization



- The enthalpy of a liquid is lower than that of its vapor.

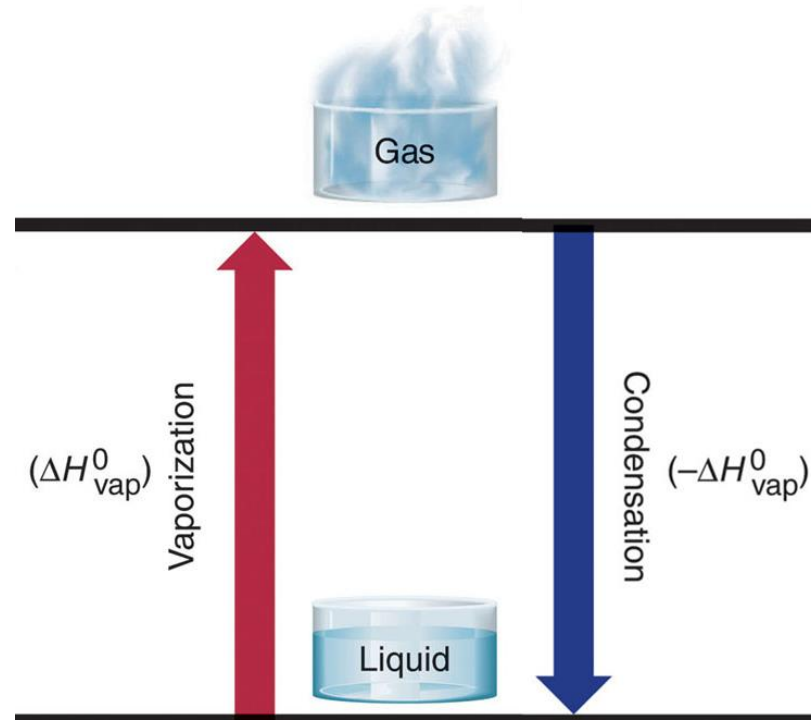
Enthalpy of Vaporization



Liquid	ΔH_{vap}
Diethyl ether, $(\text{C}_2\text{H}_5)_2\text{O}$	29.1
Methyl alcohol, CH_3OH	38.0
Ethyl alcohol, $\text{CH}_3\text{CH}_2\text{OH}$	42.6
Water, H_2O	44.0

Stronger
intermolecular
forces in liquid
water than in
liquid ether.

Enthalpy of Vaporization



Remember
Enthalpy is a STATE
Function!

$$H_{\text{vapor}} - H_{\text{liquid}} = \Delta H_{\text{vaporization}}$$

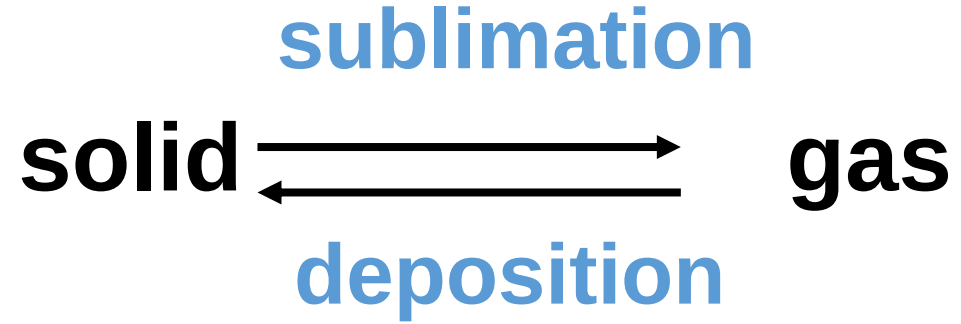
$$H_{\text{liquid}} - H_{\text{vapor}} = \Delta H_{\text{condensation}}$$

$$\Delta H_{\text{condensation}} = -\Delta H_{\text{vaporization}}$$

+ve

-ve

Phase Changes: Solid to Gas



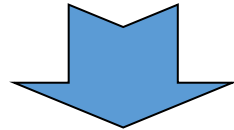
- Solids and gasses can interconvert directly, without ever becoming liquid.



Enthalpy of Sublimation

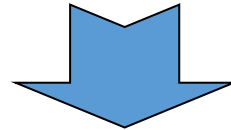
$$\Delta H_{\text{sub}} = \Delta H_{\text{fus}} + \Delta H_{\text{vap}}$$

solid



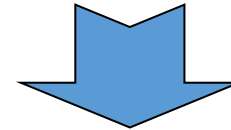
gas

solid

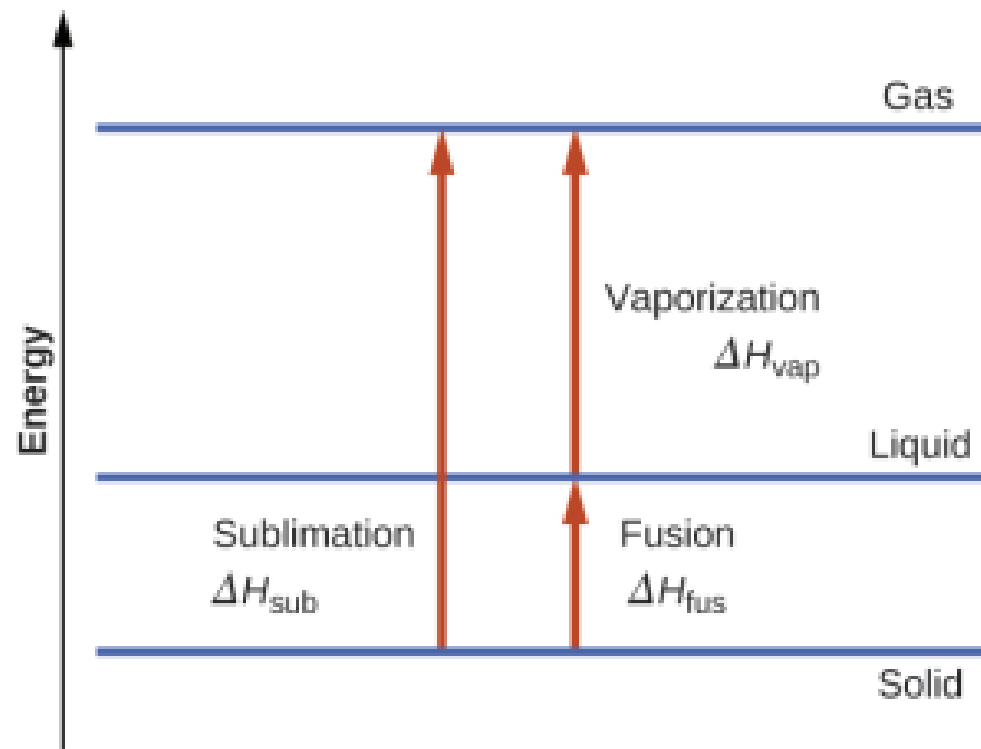
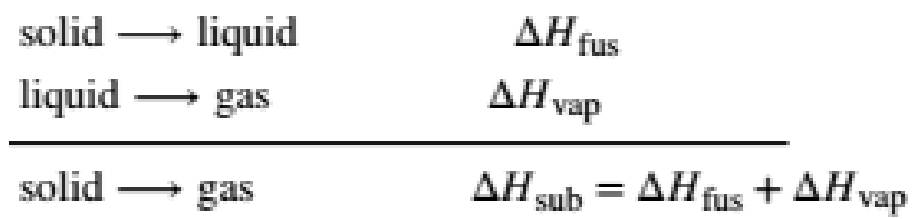


liquid

liquid



gas



Quantitative and Qualitative Aspects of Phase Changes

Heat Involved in Phase Changes

- ***Within a phase***, heat flow is accompanied by a ***change in temperature***, since the average E_k of the particles changes.

$$q = (\text{amount}) \times (\text{heat capacity}) \times \Delta T$$

- ***During a phase change***, heat flow occurs at ***constant temperature***, as the average distance between particles changes.

$$q = (\text{amount})(\Delta H \text{ of phase change})$$

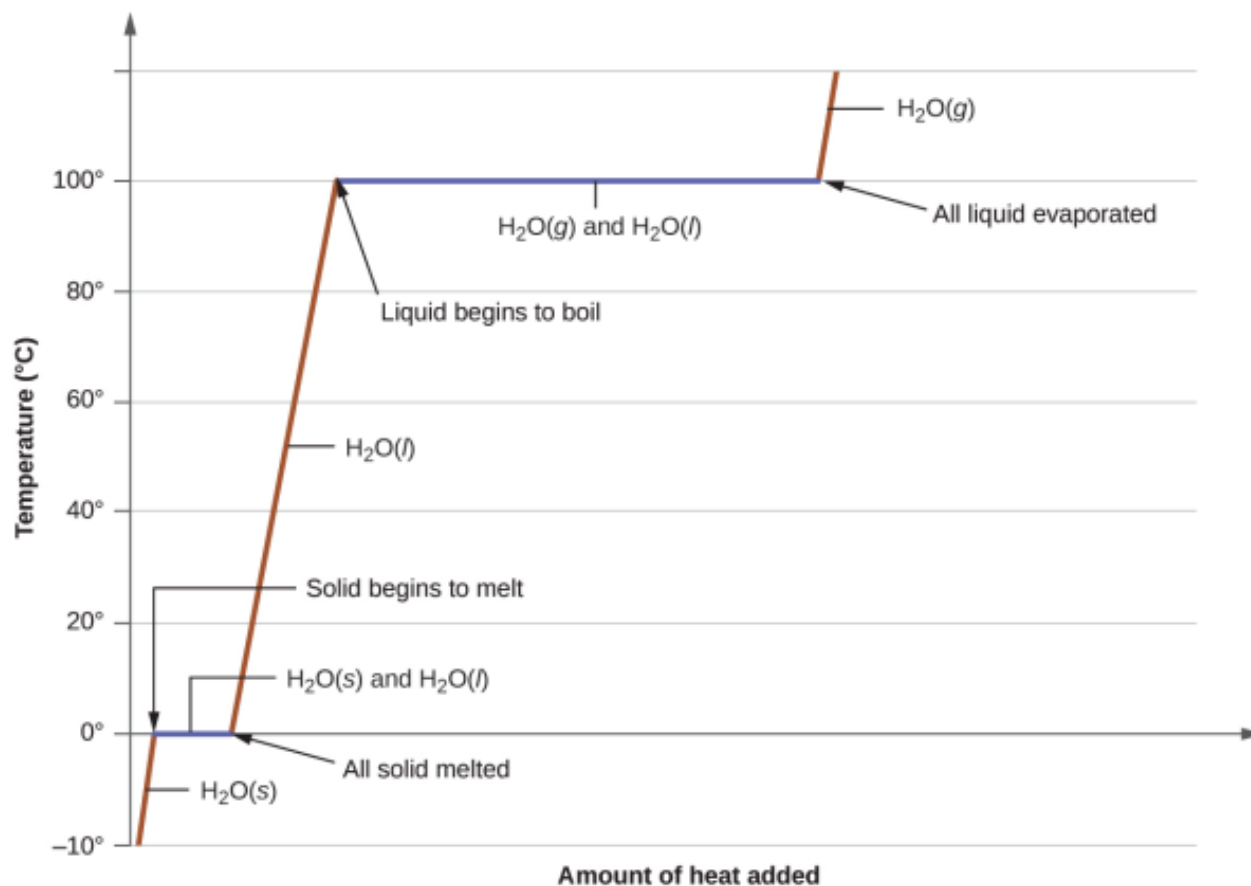


Figure 10.29 A typical heating curve for a substance depicts changes in temperature that result as the substance absorbs increasing amounts of heat. Plateaus in the curve (regions of constant temperature) are exhibited when the substance undergoes phase transitions.